

16.1 Vector Fields

1 Warm Up

Problem 1: Find the gradient of $f(x, y) = x^2 + e^{xy}$.

2 Motivation

We switch our attention to a new definition called vector fields. As of right now we take the definition as is, as we develop some intuition for what these are and how they could be useful!

3 Definitions & Theorems

1. Our story of vector fields, of course begins with the definition of one. A vector field on some set $D \subseteq \mathbb{R}^2$ is a function F which assigns to each point (x, y) a 2-dimensional vector $F(x, y)$.
2. A vector field on some set $E \subseteq \mathbb{R}^3$ is a function F which assigns each point (x, y, z) a 3-dimensional vector $F(x, y, z)$.
3. If we remember back from Chapter 14, the gradient $\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle$. Hence the gradient is really a vector field over $\mathbb{R}^2$ and hence we call it a gradient vector field. If $F(x, y)$ is the gradient of some function $f(x, y)$, we call F a conservative vector field.
4. For right now our sole objective is to get some intuition towards what vector fields look like and how we can visualize them.

4 Examples

Problem 1. Draw the following vector fields

1. $F(x, y) = 2\hat{i}$
2. $F(x, y) = -x\hat{i} - y\hat{j}$
3. $F(x, y) = y\hat{i} - x\hat{j}$
4. $F(x, y) = x^2\hat{i} + y^2\hat{j}$

$$5. F(x, y) = y^2 \hat{i} + x^2 \hat{j}$$

$$6. F(x, y) = \frac{x}{\sqrt{x^2+y^2}} \hat{i} + \frac{y}{\sqrt{x^2+y^2}} \hat{j}$$

$$7. F(x, y) = \hat{i} + x \hat{j}$$

$$8. F(x, y) = x \hat{i} + \hat{j}$$

5 Scratch Work